

Generative Adversarial Network–based Noncontrast CT Angiography for Aorta and Carotid Arteries

Jinhao Lyu, MD* • Ying Fu, PhD* • Mingliang Yang, MS • Yongqin Xiong, MD • Qi Duan, MS • Caohui Duan, PhD • Xueyang Wang, MD • Xinbo Xing, MS • Dong Zhang, PhD • Jiayi Lin, PhD • Chuncai Luo, MS • Xiaoxiao Ma, MS • Xiangbing Bian, MS • Jianxing Hu, MS • Chenxi Li, BS • Jiayu Huang, BS • Wei Zhang, BS • Yue Zhang, MS • Sulian Su, BS • Xin Lou, MD

From the Department of Radiology, Chinese PLA General Hospital, 28 Fuxing Rd, Haidian District, Beijing 100853, China (J. Lyu, Y.X., Q.D., C.D., X.W., X.X., D.Z., J. Lin, C. Luo, X.M., X.B., J. Hu, C. Li, J. Huang, X.L.); College of Medical Technology, School of Computer Science and Technology, Beijing Institute of Technology, Beijing, China (Y.F., M.Y., X.L.); Department of Radiology, Brain Hospital of Hunan Province, Changsha, China (W.Z.); Department of Radiology, Xiangyang No. 1 People's Hospital, Hubei University of Medicine, Xiangyang, China (Y.Z.); and Department of Radiology, Xiamen Humanity Hospital, Xiamen, China (S.S.). Received March 17, 2023; revision requested May 15; final revision received September 28; accepted October 4. **Address correspondence to** X.L. (email: lowxin@301hospital.com.cn). Supported by the National Natural Science Foundation of China (grants 81825012, 82151309, 82271952, 62331006, 62171038, and 62088101).

*J. Lyu and Y.F. contributed equally to this work.

Conflicts of interest are listed at the end of this article.

See also the editorial by Zhang and Turkbey in this issue.

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Background: Iodinated contrast agents (ICAs), which are widely used in CT angiography (CTA), may cause adverse effects in humans, and their use is time-consuming and costly.

Purpose: To develop an ICA-free deep learning imaging model for synthesizing CTA-like images and to assess quantitative and qualitative image quality as well as the diagnostic accuracy of synthetic CTA (Syn-CTA) images.

Materials and Methods: A generative adversarial network (GAN)–based CTA imaging model was trained, validated, and tested on retrospectively collected pairs of noncontrast CT and CTA images of the neck and abdomen from January 2017 to June 2022, and further validated on an external data set. Syn-CTA image quality was evaluated using quantitative metrics. In addition, two senior radiologists scored the visual quality on a three-point scale (3 = good) and determined the vascular diagnosis. The validity of Syn-CTA images was evaluated by comparing the visual quality scores and diagnostic accuracy of aortic and carotid artery disease between Syn-CTA and real CTA scans.

Results: CT scans from 1749 patients (median age, 60 years [IQR, 50–68 years]; 1057 male patients) were included in the internal data set: 1137 for training, 400 for validation, and 212 for testing. The external validation set comprised CT scans from 42 patients (median age, 67 years [IQR, 59–74 years]; 37 male patients). Syn-CTA images had high similarity to real CTA images (normalized mean absolute error, 0.011 and 0.013 for internal and external test set, respectively; peak signal-to-noise ratio, 32.07 dB and 31.58 dB; structural similarity, 0.919 and 0.906). The visual quality of Syn-CTA and real CTA images was comparable (internal test set, $P = .35$; external validation set, $P > .99$). Syn-CTA showed reasonable to good diagnostic accuracy for vascular diseases (internal test set: accuracy = 94%, macro F1 score = 91%; external validation set: accuracy = 86%, macro F1 score = 83%).

Conclusion: A GAN-based model that synthesizes neck and abdominal CTA-like images without the use of ICAs shows promise in vascular diagnosis compared with real CTA images.

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CT angiography (CTA) is a noninvasive and effective vascular imaging technique widely used for the diagnosis of vascular lesions, such as aortic aneurysms and dissections (1). However, CTA necessitates the intravascular administration of iodinated contrast agents (ICAs), which pose risks to patients with iodine allergy, renal insufficiency, or multiple myeloma (2,3). Additionally, ICA-based CTA is expensive and time-consuming, and studies have shown that repeated administration of ICAs could increase the risk of adverse reactions (4,5). Therefore, it is promising to develop a method to bypass ICAs and acquire CTA-like images from noncontrast CT (NCCT) images.

Intravenous injection of ICAs during CTA increases the differential attenuation of blood and surrounding

tissues on scans. Therefore, highlighting small differences in the appearance of different tissues on NCCT images is critical for obtaining CTA-like images from NCCT. Deep learning networks can distinguish class values with small differences using a nonlinear combination of quadratic operations (6–8), which provides a technical direction to obtain CTA-like images free of ICAs.

Preliminary achievements in deep learning–based medical image translation studies include MRI enhancement (9,10), low-dose CT enhancement (11,12), and low-dose contrast CT enhancement (13). To the best of our knowledge, no studies obtaining CTA-like images independently of ICAs have been reported. Synthesizing

Abbreviations

CTA = CT angiography, CTA-GAN = GAN-based CTA imaging, GAN = generative adversarial network, ICA = iodinated contrast agent, NCCT = noncontrast CT, Syn-CTA = synthetic CTA

Summary

Generative adversarial network–based CT angiography (CTA) images of the neck or abdomen obtained without iodinated contrast agents had image quality comparable to that of real CTA images, and reasonable to good diagnostic accuracy.

Key Results

- A synthetic noncontrast CT angiography (CTA) model produced images with carotid and aortic visual quality comparable to that of real CTA scans.
- The model-generated CTA-like images had a diagnostic accuracy for vascular pathology of 86%, with a macro F1 score of 83%, for the external validation set.

CTA-like images from NCCT images based on deep learning networks and using these images in clinical diagnosis remain a challenge. For example, traditional deep learning models do not sufficiently solve the mapping translation problem of paired misaligned images. Furthermore, previous studies of medical image translation focus on a single anatomic location, whereas clinical diagnosis is frequently performed in multiple locations (14,15).

The aim of this study was to develop a CTA imaging model based on a generative adversarial network (GAN) (16–21) to synthesize high-quality CTA-like images independently of ICAs and to assess the feasibility of using these synthetic CTA (Syn-CTA) images in aiding clinical diagnosis. Internal and external test data were used to evaluate Syn-CTA images in terms of quantitative metrics, visual quality, and diagnostic accuracy for vascular diseases.

Materials and Methods

Data Acquisition and Processing

This retrospective study was approved by the institutional review board and ethics committee of Chinese PLA General Hospital (ClinicalTrials.gov: NCT05471869). The requirement for informed consent was waived because the images were acquired during daily practice. This study included neck and abdominal paired NCCT-CTA data collected at Chinese PLA General Hospital from January 2017 to December 2022. Additionally, external data of paired abdominal NCCT-CTA scans collected from June 2018 to March 2023 at the Brain Hospital of Hunan Province were included (Fig 1). These data were obtained using GE, Siemens, or Philips equipment and saved in Digital Imaging and Communications in Medicine, or DICOM, format. Each NCCT and CTA scan (except for scans with a thickness of 5 mm) was resampled into $0.67 \times 0.67 \times 1.25$ -mm³ volumes and consisted of 75–490 sections with a matrix size of 512×512 pixels.

The contrast agent used for all CTA examinations was iodine, with a concentration of 370 mg/mL and an injection rate of 4.5 mL/sec. Each volume was processed independently by normalizing the intensities in the range –2000 to 2095 pixel value to –1 to 1. Scans with poor image quality upon manual inspection were excluded according to the inclusion and exclusion criteria (Appendix S1). The final obtained data were randomly divided (with scan as the basic unit) among the training, validation, and test sets for model development.

Model Development

The developed GAN-based CTA imaging (CTA-GAN) model (Fig 2) included a generator (21), a corrector (21,22), and a discriminator (23). The generator was designed for CTA-like

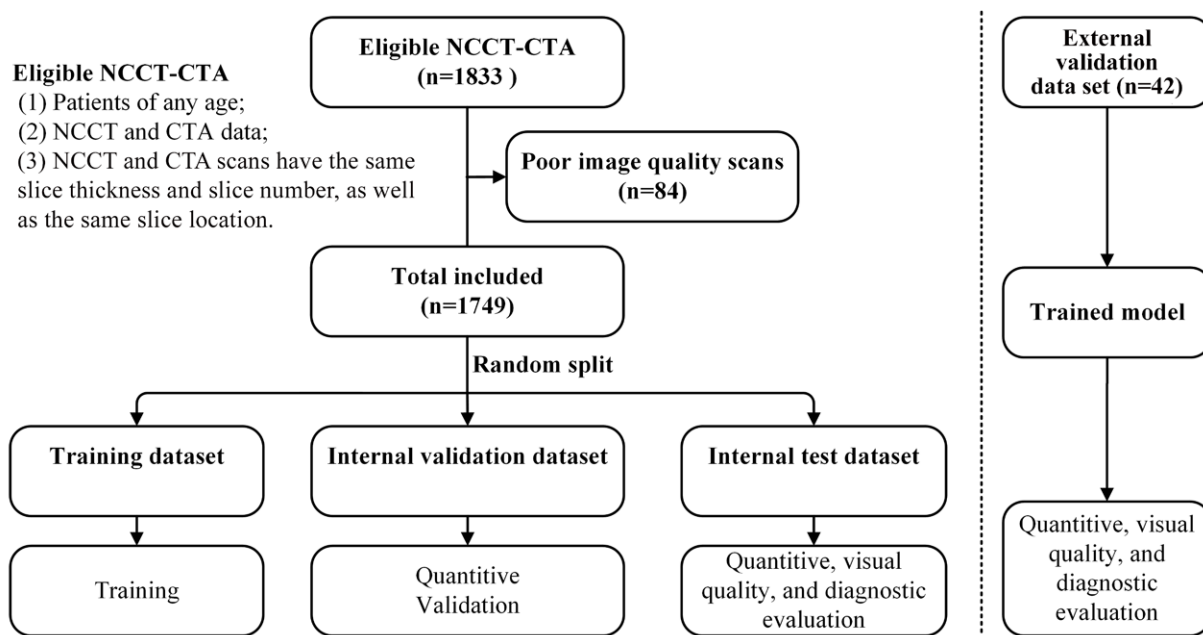


Figure 1: Flowchart of paired noncontrast (NCCT) and CT angiography (CTA) scans included in the study.

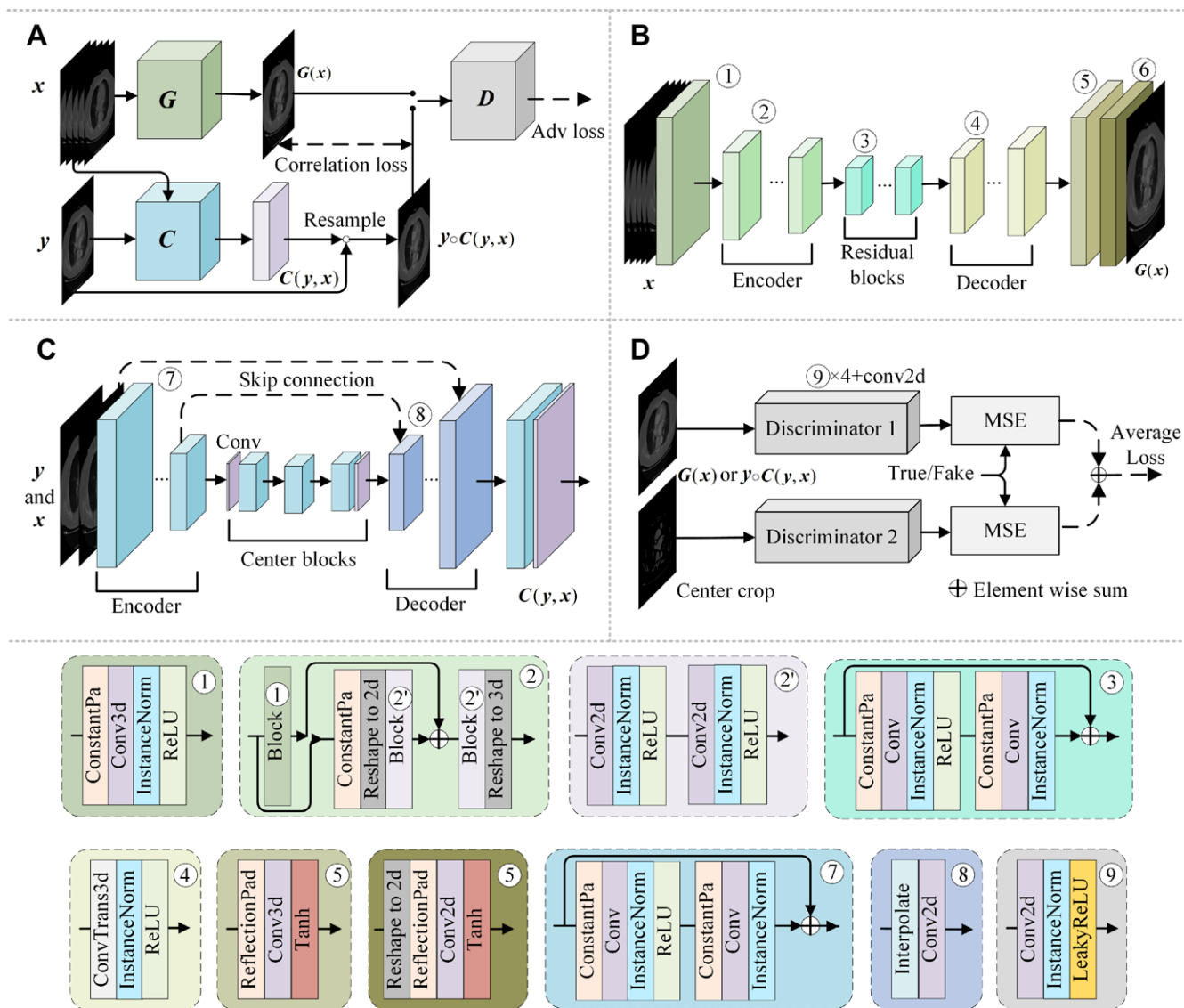


Figure 2: Network architecture of the generative adversarial network-based CT angiography (CTA) imaging (CTA-GAN) model. Diagrams show the (A) overall architecture, (B) generator architecture, (C) corrector architecture, and (D) discriminator architecture of the CTA-GAN model. Note that x denotes the input of the generator, that is, the normalized noncontrast CT scan; y denotes the target output of the generator; $G(x)$ denotes the normalized synthetic CTA scan produced by the generator; and $C(y, x)$ denotes the correction matrix. Modules 1–9 at the bottom are the basic components of the model shown in A–D. Adv = adversarial, Conv = convolution, InstanceNorm = instance normalization, MSE = mean square error, ReLU = rectified linear unit, Tanh = hyperbolic tangent.

image synthesis based on NCCT images. The corrector was used to adapt the real CTA image to the Syn-CTA space so that the error loss could be calculated more accurately. The discriminator estimated the probability of the input images being real CTA images versus Syn-CTA images. Software and details of the CTA-GAN model are provided in Appendix S1. The platform used for this experiment was a Linux system server with four NVIDIA GeForce RTX 3090 Ti graphics processing units and 256 GB of RAM. The source code for this study is available at <https://github.com/ying-fu/CTA-GAN>. Details of the model training are available in Appendix S1.

Quantitative Evaluation

The results of the proposed CTA-GAN model were compared quantitatively with those of two other image translation

models, pix2pix (16) and RegGAN (21). Normal mean absolute error, peak signal-to-noise ratio, and the structural similarity index measure were used to evaluate the similarity between real CTA and Syn-CTA images (21,24). Details can be found in Appendix S1.

Visual Quality Evaluation

Image quality affects how quickly and accurately a radiologist can diagnose a disease. Two senior radiologists (board-certified radiologists J. Lyu and X.M., both with 10 years of experience) independently assessed the image quality of the CTA and Syn-CTA images. Any disagreement was resolved by consensus reading. The radiologists assessed the image quality of the Syn-CTA and real CTA scans using a subjective three-point scale (visual quality score) (25,26), as follows: 1, poor quality;

2, satisfactory quality; and 3, good quality. Specifically, image quality assessment included the subcategories vessel wall clarity, lumen edge sharpness, and lumen wall contrast (Appendix S1, Fig S1).

Diagnostic Evaluation

The Syn-CTA images and the real CTA images were anonymized for each scan and were then randomized and presented by series number to the same two radiologists who performed the visual quality evaluation for independent reading. The vascular diagnosis (aneurysm, dissection, atherosclerosis, or healthy arteries) based on each scan was determined by the two radiologists. Any diagnosis disagreement was resolved by consensus reading (Appendix S1). The vascular diagnosis read from the real CTA images was treated as the ground truth.

Statistical Analysis

The visual quality scores of the Syn-CTA and real CTA images were compared using the χ^2 test or Fisher exact test. Specifically, in this study, the Fisher exact test was used when the theoretical frequency T was less than 5; otherwise, the χ^2 test was employed (27). In addition, visual quality scores for real CTA and Syn-CTA images were dichotomized (poor or satisfactory quality [score = 1 or 2] vs high quality [score = 3]) for comparison. For all analyses, $P < .05$ was considered indicative of a statistically significant difference.

The reliability of the vascular diagnosis was assessed using a confusion matrix (28). The accuracy rate was used to evaluate the model output for each diagnosis category as well as the overall accuracy of the model. The overall performance of the CTA-GAN model was evaluated using the F1 score metric. F1 score values range from 0 to 1, with 1 representing perfect precision and recall by the model and 0 representing the worst possible output of the model (Appendix S1).

A prior study (29) found that the prevalence of dissection and aneurysm at CTA was 8%. The sample size was determined using the diagnostic testing equation comparing sensitivity (specificity) of a test to a prespecified value (30). To ensure that the type I error rate would not exceed 5% and the type II error rate would not exceed 20%, we estimated the minimum sample size with an anticipated sensitivity of 95% and specificity of 90% for the Syn-CTA images by selecting a prespecified value of 50% for the sensitivity and specificity, thus estimating a required minimum of 88 cases for the testing group. As per the study design, the testing group constituted 10% of all sample cases. Hence, given a dropout rate of 0%, 880 was the calculated minimum sample size for the entire study.

Results

Patient Characteristics

Of the 1833 eligible patients, 84 patients with poor image quality were excluded, and 1749 patients (median age, 60 years [IQR, 50–68 years]; 1057 [60.4%] male patients and 692 [39.6%] female patients) were included in the analysis. CT scans from 1137 patients were used for model training; scans from 400 patients were used for model development validation; and scans

from 212 patients were used for model testing (Fig 1). Table 1 shows details of patient demographics and scan parameters. In addition, the external independent validation set comprised 42 patients (median age, 67 years [IQR, 59–74 years]; 37 [88.1%] male patients and five [11.9%] female patients).

Quantitative Evaluation

In the quantitative evaluation of the test set, the CTA-GAN model performed the best among the three models, and pix2pix performed the worst. The models had good agreement on both the validation and test sets. Overall, the normalized mean absolute error for Syn-CTA images generated by CTA-GAN was 0.011, the mean peak signal-to-noise ratio was 32.07 dB, and the mean structural similarity index measure was 0.919 (Table 2). In the external validation set, the normalized mean absolute error for Syn-CTA images generated by CTA-GAN was 0.013, the mean peak signal-to-noise ratio was 31.58 dB, and the mean structural similarity index measure was 0.906.

Visual Quality Evaluation

The rate of high-quality scores (score = 3) in the Syn-CTA test set was greater than 90% for all subcategories of the image quality assessment, and the overall rate of a high-quality score was 95% (Fig 3). Accordingly, there was no evidence of a difference ($P = .35$ for internal test set, $P > .99$ for external validation set) between the overall scores of Syn-CTA images and real CTA images for image quality.

Comparisons between quantitative scores and visual scores were performed to explore the relationship between quantitative metrics of the proposed model and clinical practice. The CTA-GAN model obtained the best metrics among all of the methods evaluated, and the quantitative score was positively correlated with the visual score. Representative cases are shown in Figures 4 and 5; more examples are shown in Figures S2 and S3.

Diagnostic Evaluation

A total of 212 scans from patients at Chinese PLA General Hospital were evaluated in the clinical setting to compare diagnostic performance with that of real CTA images. The overall accuracy rate of the synthetic images was 94% (200 of 212). The overall sensitivity of the synthetic images was 93%, with a sensitivity of 87% (20 of 23) for aneurysms, 90% (nine of 10) for dissections, 94% (130 of 138) for atherosclerosis, and 100% (41 of 41) for healthy arteries. The overall specificity of the synthetic images was 98%, with a specificity of 100% (189 of 189) for aneurysms, 99% (200 of 202) for dissections, 99% (73 of 74) for atherosclerosis, and 95% (162 of 171) for healthy arteries (Table 3). The macro F1 score for the internal test set was 91%.

Syn-CTA and real CTA images from 42 patients at the Brain Hospital of Hunan Province were used as the external validation set for evaluating diagnostic performance. The overall accuracy rate of the synthetic images was 86% (36 of 42). The overall sensitivity of the synthetic images was 88%, with a sensitivity of 100% (12 of 12) for aneurysms, 67% (eight of 12) for dissections, 87% (13 of 15) for atherosclerosis, and 100% (three of three)

Table 1: Scan Parameters and Patient Demographics

Parameter	Neck CT*	Abdominal CT†		
		GE Equipment	Siemens Equipment	Philips Equipment
Scan parameters				
X-ray tube voltage (kVp)	100	120	120	100
X-ray tube current (mA)‡	122.5 (90–270)	271 (200–370)	252 (217.5–306)	562 (401–689)
Section thickness (mm)	0.625	0.625	1.000	1.500
Pitch	0.992	0.992	0.600	0.925
Field of view (cm)	25 × 25	42.9 × 42.9	38 × 38	35 × 35
Reconstruction algorithm	ASiR-V	ASiR-V	SAFIRE	iDose4
No. of detector rows	256	256	64	128
Rotation time (sec)	0.5	0.5	0.5	0.4
Internal set				
No. of patients	908	364	440	37
Age (y)‡	59 (50–67)	61 (48–68)	59 (51–68)	55 (48–65)
M:F ratio	376:532	225:139	274:166	20:17
Diagnosis category ratio§	2:0:575:331	30:20:226:88	40:20:249:131	0:1:25:11
External set				
No. of patients	0	4	0	38
Age (y)‡	NA	57.5 (52.5–64)	NA	67.5 (52.5–64)
M:F ratio	NA	4:0	NA	33:5
Diagnosis category ratio§	NA	0:2:2:0	NA	12:10:13:3

Note.—NA = not applicable.

* Neck CT was performed with GE equipment; CT scans covered at minimum the frontal sinus to the aortic arch.

† Abdominal CT scans covered at minimum the diaphragm to the iliofemoral arteries. Thorax and abdominal CT angiography scans were included in this category.

‡ Data are medians, with IQRs in parentheses.

§ Diagnosis categories are aneurysm:dissection:atherosclerosis:healthy arteries. The ratios for abdominal CT for all equipment types combined are as follows: internal set, 69:40:500:232; external set, 12:12:15:3.

Table 2: Quantitative Evaluation of CT Angiography Images Synthesized by Different Models

Model	Internal Validation Set			Internal Test Set			External Validation Set		
	NMAE	PSNR	SSIM	NMAE	PSNR	SSIM	NMAE	PSNR	SSIM
pix2pix	0.026	25.30	0.843	0.026	25.19	0.797	0.026	24.96	0.784
RegGAN	0.013	31.05	0.924	0.014	30.10	0.890	0.015	29.86	0.883
CTA-GAN	0.010	33.76	0.948	0.011	32.07	0.919	0.013	31.58	0.906

Note.—Lower values indicate better performance for NMAE, whereas higher values indicate better performance for PSNR and SSIM. CTA-GAN = generative adversarial network–based CT angiography imaging, NMAE = normalized mean absolute error, PSNR = peak signal-to-noise ratio, SSIM = structural similarity index measure.

for healthy arteries. The overall specificity of the synthetic images was 95%, with a specificity of 97% (29 of 30) for aneurysms, 100% (30 of 30) for dissections, 89% (24 of 27) for atherosclerosis, and 95% (37 of 39) for healthy arteries (Table 3). The macro F1 score for the external validation set was 83%.

In the internal test set, the overall rate of incorrect diagnosis was 6% (12 of 212); one of 10 dissections was missed, and three of 20 aneurysms were misdiagnosed. In the external validation test set, the overall rate of incorrect diagnosis was 14% (six of 42); four of 12 dissections were missed, while all aneurysms were correctly diagnosed. The CTA-GAN model encountered difficulties with certain contrast enhancements (Fig S4).

Discussion

In this work, we developed a generative adversarial network (GAN)–based imaging model for the neck and abdomen that generates CT angiography (CTA)–like images from noncontrast CT images. In the quantitative image quality evaluation and visual quality evaluation, the synthetic CTA (Syn-CTA) images generated by the GAN-based CTA imaging model were comparable to real CTA scans. Diagnosis based on Syn-CTA images showed reasonable to good diagnostic accuracy for vascular diseases (internal test set: accuracy = 94%, macro F1 score = 91%; external validation set: accuracy = 86%, macro F1 score = 83%). In the external validation set, the sensitivity was 100% (12 of 12)

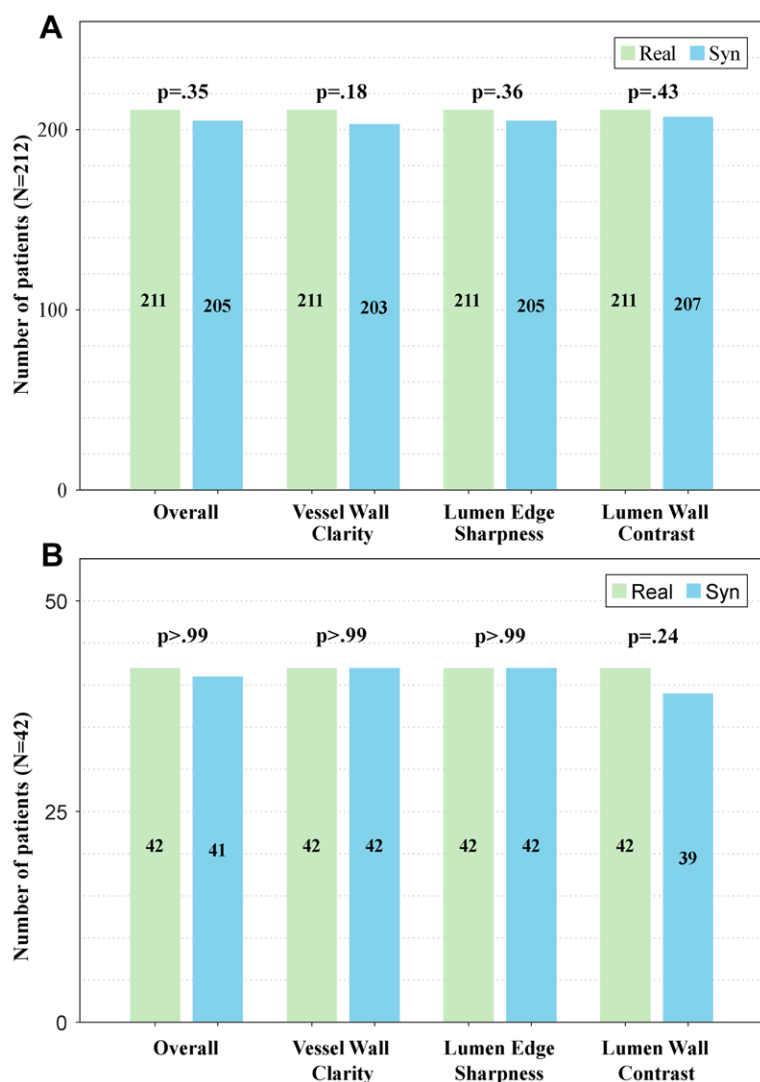


Figure 3: Comparison of visual quality scores between synthetic (Syn) and real CT angiography (CTA) images in the (A) internal test set and (B) external validation set. Graphs show the number of patients with a high-quality score (score = 3) for overall image quality and for subcategories of the image quality assessment. There was no evidence of a difference in image quality between synthetic and real CTA images ($P = .35$ for internal test set, $P > .99$ for external validation set).

for aneurysms, 67% (eight of 12) for dissections, 87% (13 of 15) for atherosclerosis, and 100% (three of three) for healthy arteries. This work demonstrates that noncontrast artificial intelligence technology could become an important complement to iodine-based CTA imaging.

In clinical diagnosis, NCCT cannot be used for a definitive diagnosis by itself and only occasionally provides suggestive information on vascular lesions to the physician. CTA is needed for the definitive diagnosis of vascular diseases. In our study, we developed a noncontrast alternative to CTA using a GAN-based synthetic model. The vascular locations, vascular enhancement, and structural similarities between Syn-CTA and real CTA images were remarkable. In this study, Syn-CTA images were generated with a CTA-GAN model that uses pixels and contours as conditional information for target image generation. The neural network is sensitive to small pixel differences, allowing it to detect and highlight areas of

blood on NCCT scans that might not be distinguishable by human eyes (details can be seen in Fig S5). This mechanism forms the basis for the CTA-GAN model, enabling contrast-free highlighting of carotid and aortic vessels based on NCCT scans. While there were small differences between the Syn-CTA and real CTA images, diagnoses of aneurysm, dissection, or atherosclerosis could be aided with early screening using Syn-CTA findings, especially for patients with iodine allergy or limited access to health care.

To our knowledge, studies on the use of NCCT images for CTA-like image synthesis and clinical diagnosis completely independently of ICAs have not yet been reported. Most of the current studies focus only on the enhancement of specific quantitative indicators, leaving a “usage gap” between modeling methods and clinical applications (28,29,31,32). In the quantitative image quality evaluation, the Syn-CTA images from CTA-GAN achieved better results for normalized mean absolute error, peak signal-to-noise ratio, and structural similarity index measure than the synthetic images from the comparison models (pix2pix and Reg-GAN). In the visual quality assessment of the test data set by two radiologists, there was no evidence of a difference between the Syn-CTA and real CTA images. In addition, promising accuracy results were obtained in the diagnostic evaluation of Syn-CTA images compared to real CTA images. This work confirms that deep learning models have the potential to generate high-quality CTA-like images without contrast agent.

Our work has some inevitable limitations. First, the CTA-GAN model was developed based on data from a single center, and the trained model sometimes had errors or omissions in the enhanced region, such as failing to display the dissection flap in some cases. The vessel wall and intimal flap manifestation of dissection can be complex, which

poses pressing challenges in clinical diagnosis as well as in deep learning model training. In subsequent research, a large sample of multicenter data will be collected to update the model, aiming to improve the accuracy and generalizability of CTA-GAN. Second, although we included processes such as correction modules and multifield discriminators in the training of the CTA-GAN model, these do not completely solve the mode collapse and training instability problems of the GAN, and individually generated images can still have some omitted enhancement or incorrect highlighting problems. In future model construction, we will try to address these issues by further combining GAN with diffusion models to train different classes of data in a balanced manner.

In addition, CTA-GAN faces some technical challenges in simultaneously synthesizing CTA-like images of large arteries, small visceral arteries, and organs for clinical assisted-diagnosis applications. First, arteries account for only a small proportion

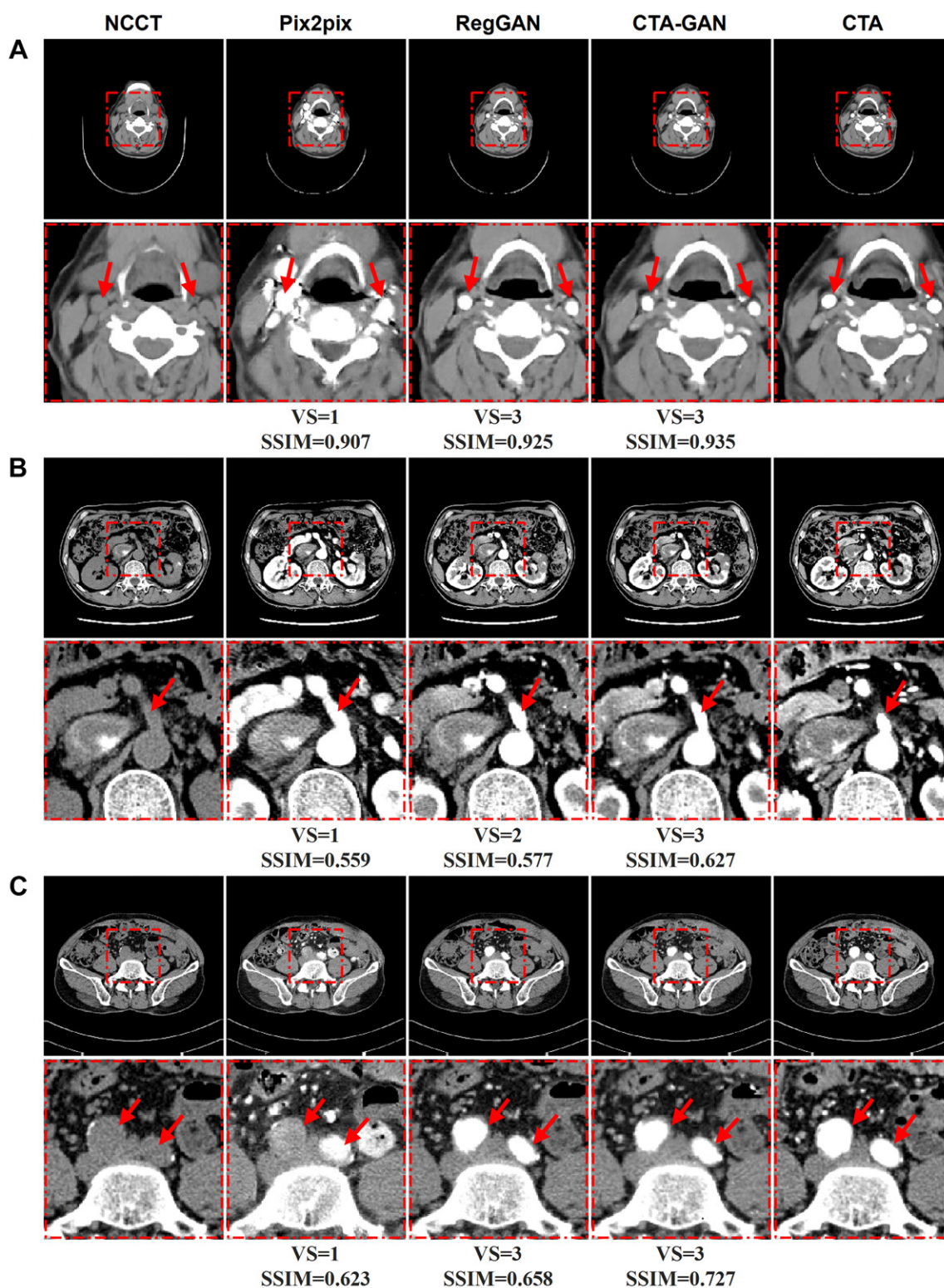


Figure 4: Comparison of noncontrast CT (NCCT) images, real CT angiography (CTA) images, and synthetic CTA images from a generative adversarial network-based CTA imaging (CTA-GAN) model and two other models (pix2pix and RegGAN) in healthy patients in the internal test set. Each pair of rows represents the same axial section, with the real CTA image (right) shown as a reference. The region within the red box in the images in the top row of each example is enlarged and shown in the bottom row. **(A)** Carotid artery images in a 43-year-old female patient. **(B)** Abdominal aorta images in a 71-year-old male patient. **(C)** Abdominal aorta images in a 57-year-old male patient. The vascular structures are blurred, and there is no attenuation difference between the vascular and surrounding tissues along most of the vascular segments on NCCT images, while the vascular location and structural similarities between the synthetic CTA-GAN and real CTA images are striking. The synthesized CTA images generated by the CTA-GAN model highlight arterial blood more accurately than the images generated by the other two models. Arrows indicate target vascular structures. SSIM = structural similarity index measure, VS = visual score.

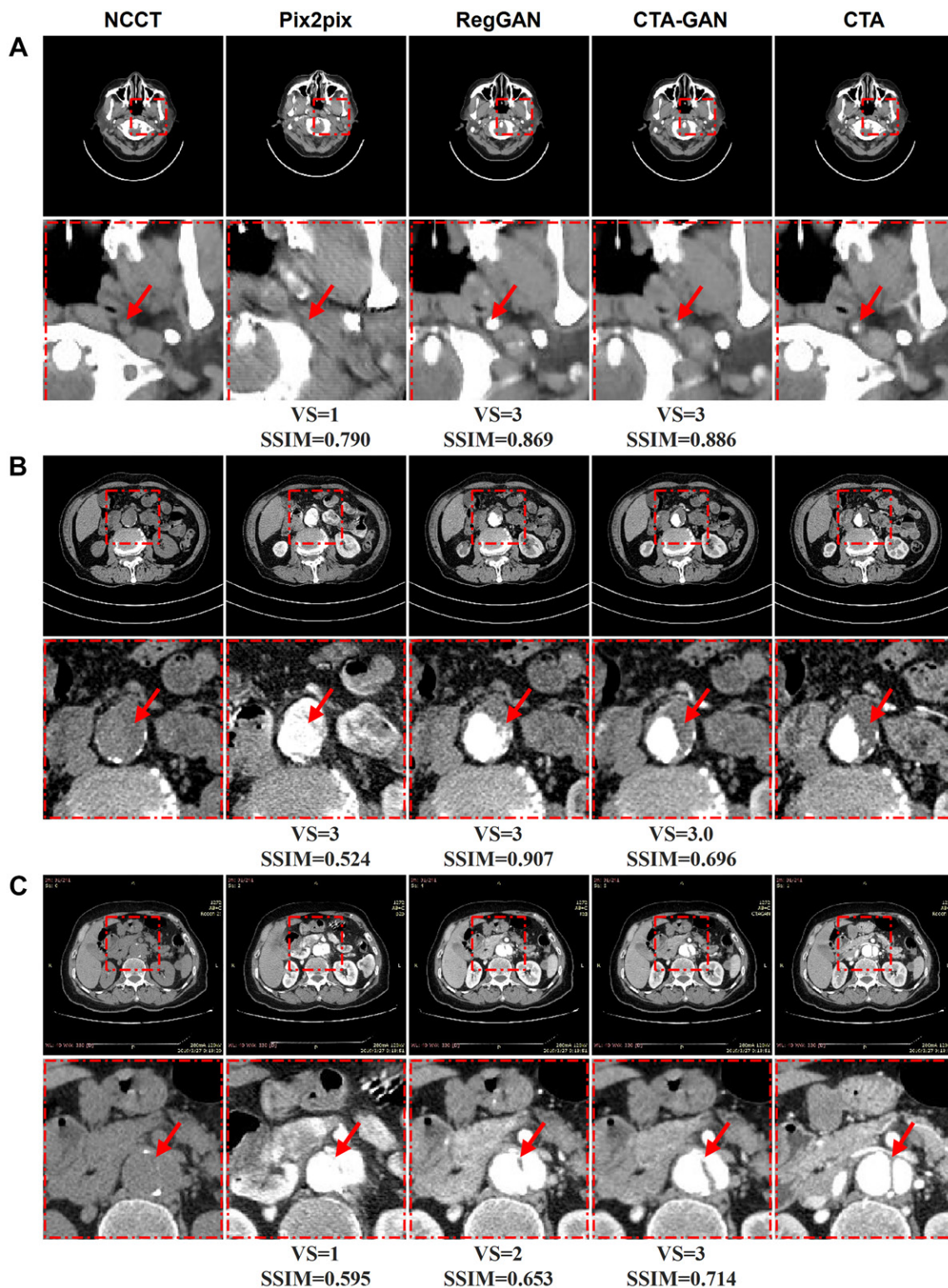


Figure 5: Comparison of noncontrast CT (NCCT) images, real CT angiography (CTA) images, and synthetic CTA images from a generative adversarial network–based CTA imaging (CTA-GAN) model and two other models (pix2pix and RegGAN) in patients with vascular diseases in the internal test set. Each pair of rows represents the same axial section, with the real CTA image (right) shown as a reference. The region within the red box in the images in the top row of each example is enlarged and shown in the bottom row. **(A)** Images in a 54-year-old male patient with internal carotid artery stenosis show substantial internal carotid artery narrowing (arrows). **(B)** Images in a 74-year-old male patient with abdominal aortic aneurysm show aneurysmal dilation of the abdominal aorta with remarkable wall thickening (arrows), which suggests a mural thrombus. **(C)** Images in a 62-year-old male patient with abdominal aortic dissection show no calcified intimal lines, but a few calcified deposits in the vessel wall (arrows) and unclear vascular status on the NCCT image. The synthesized CTA images generated by the CTA-GAN model highlight arterial blood more accurately than the images generated by the other two models. SSIM = structural similarity index measure, VS = visual score.

Table 3: Confusion Matrix of Aortic and Carotid Artery Diagnosis by Radiologists Using Syn-CTA versus Real CTA Images in Internal Test Set and External Validation Set

	Diagnosis Based on Real CTA Images				
Diagnosis Based on Syn-CTA Images	Aneurysm	Dissection	Atherosclerosis	Healthy Arteries	Total
Internal test set					
Aneurysm	20	0	0	0	20
Dissection	1	9	1	0	11
Atherosclerosis	0	1	130	0	131
Healthy	2	0	7	41	50
Total	23	10	138	41	212
External test set					
Aneurysm	12	0	1	0	13
Dissection	0	8	0	0	8
Atherosclerosis	0	3	13	0	16
Healthy	0	1	1	3	5
Total	12	12	15	3	42

Note.—Data are numbers of patients. CTA = CT angiography, Syn-CTA = synthetic CTA.

of the whole body. To synthesize CTA-like images from NCCT images, it is critical that the model have an attention mechanism focusing on vascular structures. Adding information on organs that characterize large volumes on scans will decrease the effectiveness of the attention mechanism. Second, paired image registration for small branches and organs is a difficult task with current technology.

In conclusion, we propose a generative adversarial network (GAN)-based CT angiography (CTA) model that uses noncontrast CT images of the neck and abdomen to synthesize CTA-like images. GAN-based CTA-like images of the neck or abdomen without iodinated contrast agent had image quality comparable to that of real CTA images, and reasonable to good diagnostic accuracy. With further studies and assessment, this model may provide a fast and low-cost auxiliary CTA-like imaging method for patients, especially those with iodine allergy or limited access to health care. The model has the potential to reduce the occurrence of adverse events related to contrast agents.

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Data sharing: Data generated or analyzed during the study are available from the corresponding author by request.

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